UE26CY151A Engineering Chemistry · Unit 2 · Electrochemistry and Corrosion

Important points

Definitions, statements, key points, "why" reasoning and exam traps, topic by topic.

Class 001 · Origin of electrode potential, cell potential · compiled 17 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S001·10	Deck 001, slide 10 (rendered .pptx)
T001 00:08:22	AV-summary transcript 001, segment start
ESA Jan-25 s2 2b	Past paper: Jan 2025 set 2, question 2.b
ESA Jul-23a 4d	Past paper: Jul 2023 (first UE22 paper), question 4.d

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1 Electrochemistry and the two types of cell

Definitions

Electrochemistry. The branch of chemistry that deals with the interconversion of chemical energy and electrical energy. S001·3 · T001 00:00:45

Galvanic (voltaic) cell. A cell that converts chemical energy to electrical energy, using a spontaneous redox reaction; e.g. batteries and fuel cells. S001·3 · T001 00:01:15

Electrolytic cell. A cell that converts electrical energy to chemical energy: an external supply drives a non-spontaneous redox reaction; e.g. cells used in electroplating and electrorefining. S001·3 · T001 00:02:19

Anode / cathode. The anode is the electrode where oxidation takes place; the cathode is the electrode where reduction takes place. S001·4 · T001 00:03:30

Electrolytic conductance. Conduction through the electrolyte by the movement of ions (electrons cannot move through the electrolyte). Electrolytes may be acid, alkali or salt solutions, molten electrolytes or solid electrolytes. S001·4 · T001 00:03:49

Key points

1. A galvanic cell needs a **spontaneous** redox reaction, because no energy is supplied from outside; an electrolytic cell drives a **non-spontaneous** one with external electrical energy. ESA ★★ T001 00:01:30 · T001 00:02:34
2. Oxidation and reduction always happen simultaneously; producing electrical energy needs an exchange of electrons, hence a redox reaction. ESA ★ S001·4 · T001 00:03:16
3. Electrons cannot move through the electrolyte; only ions carry the current there. ESA ★★ T001 00:03:57
4. Solid electrolyte example: zirconia doped with Y_2O_3 has oxide-ion vacancies, so O^{2-} ions move through the solid. ESA ★ transcript only T001 00:04:58
5. Molten electrolytes work because melting (e.g. NaCl) frees the ions to move. ESA ★ transcript only T001 00:04:28

2 Electrode potential and its origin: the Helmholtz double layer

Definitions

Electrode potential (E). When a metal rod is dipped in a solution of its own ions, the electrical potential developed at the interface of the metal and its solution. It is denoted by E. S001·4 · T001 00:05:32

Helmholtz electrical double layer. The layer of charge on the metal surface together with the layer of oppositely charged ions it attracts from the solution, formed at the metal–solution interface; it is responsible for the electrode potential. S001·6 · S001·7 · T001 00:08:22

Statements in the class material

> *"When a metal M is in contact with solution containing its ions* $\mathbf{M^{n+}}$*, two reactions are possible: 1. Ionisation (oxidation)* $\mathbf{M \rightleftharpoons M^{n+} + ne^-}$*; 2. Deposition (reduction)* $\mathbf{M^{n+} + ne^- \rightleftharpoons M}$*."* S001·5

> *"Case I: if ionization is faster than deposition, the metal acquires net negative charge, consequently retards the rate of ionization and increases the rate of deposition. This ultimately leads to the establishment of equilibrium. The metal electrode gets negatively charged and attracts the layer of positive ions at the interface; an electrical double layer is formed at the interface of metal and solution known as Helmholtz electrical double layer."* S001·6

> *"Case II: if deposition is faster than ionization, the metal acquires net positive charge, consequently retards the rate of deposition and increases the rate of ionization. This ultimately leads to the establishment of equilibrium. The metal electrode gets positively charged and attracts the layer of negative ions at the interface; an electrical double layer is formed, known as Helmholtz electrical double layer."* S001·7

Key points

1. Case I: the electrons left behind make the metal negative; it attracts a layer of positive ions ($\mathbf{M^{n+}}$). Case II: the electrons taken from the metal make it positive; it attracts a layer of negative ions. ESA ★★★ S001·6 · S001·7 · ESA Jan-25 s2 2b
2. The charge on the metal is self-limiting: in case I the negative charge retards ionisation and speeds deposition until the two rates are equal (equilibrium); case II is the mirror image. ESA ★★ S001·6 · S001·7
3. Which reaction is faster depends on the nature of the metal (its tendency to oxidise or its ions to be reduced), the ion concentration, the temperature and, for gases, the pressure. ESA ★★ T001 00:10:30

Why?

Why does the double layer not keep growing?
In case I the negative charge on the metal pulls M^{n+} back (more deposition) and holds the cations in (less ionisation); the rates become equal and a dynamic equilibrium with a fixed potential difference is reached.
S001·6

Exam traps

Exam trap. In case I the metal is **negative** and the attracted layer is **positive** ions; in case II the metal is positive and the layer is negative ions. Swapping them loses the marks for ESA Jan-25 s2 2.b(i). S001·6 · S001·7 · ESA Jan-25 s2 2b

3 Standard electrode potential, electrochemical cell and cell notation

Definitions

Standard electrode potential (E°). The potential developed at the interface of metal and solution, when the metal is in contact with a solution of its own ions having unit concentration at 298 K. In case of gas electrodes the partial pressure of gas is maintained at 1 atmospheric pressure. It is represented as E°. S001·8 · T001 00:11:18

Electrochemical cell. Single electrode potentials cannot be measured, hence two electrodes are coupled together to form a cell. S001·8 · T001 00:12:06

Cell notation. A short representation of a cell: anode on the left, cathode on the right; a single line | separates two phases and a double line || is the salt bridge between the two solutions. S001·8 · T001 00:12:25

Key points

1. E° depends only on the nature of the metal, because concentration, temperature and pressure are fixed. ESA ★★ T001 00:11:56
2. In the notation the anode is written first (left), the cathode last (right). In the Daniell cell zinc is oxidised, so Zn is the anode. ESA ★★★ S001·8 · T001 00:12:55
3. Do not write Zn | $\mathrm{Zn^{2+}}$ || Cu | $\mathrm{Cu^{2+}}$: the salt bridge connects the two *solutions*, so the ions sit next to ||. ESA ★★ T001 00:13:43
4. The concentration of each ion is written in brackets after it, e.g. $\mathrm{Zn^{2+}(1\,M)}$. ESA ★★ S001·8 · T001 00:13:13

Why?

Why can a single electrode potential not be measured?
A voltmeter measures a potential *difference*; connecting it to the solution needs a second conductor in the solution, which is itself a second electrode. So two electrodes are always coupled into a cell, and E° values are measured against a reference (SHE = 0).

added for clarity S001·8 · T001 00:12:06

Exam traps

Exam trap. The anode goes on the **left**. Writing the Daniell cell as Cu | $\mathrm{Cu^{2+}}$ || $\mathrm{Zn^{2+}}$ | Zn gives $E_{cell} = -1.10$ V, a non-spontaneous cell. added for clarity S001·8 · S001·9

4 Cell potential, standard cell potential and free energy

Definitions

Cell potential (E_{cell}). The difference in electrode potentials of the electrodes constituting the cell. It is denoted by E_{cell}. S001·9 · T001 00:14:25

Standard cell potential (E°_{cell}). When all the concentrations are 1 M, all partial pressures of gases are 1 atm and temperature is 298 K, the emf is called the standard cell potential, E°_{cell}. S001·9 · T001 00:15:01

Statements in the class material

> *"$E_{cell} = E_{rhs} - E_{lhs} = E_{cathode} - E_{anode}$. E_{cell} represents the driving force for the cell reaction to take place: $\Delta G = -nFE_{cell}$. If the reaction is spontaneous ΔG is negative, thus E_{cell} should be positive; if the reaction is non-spontaneous ΔG is positive, thus E_{cell} should be negative."* S001·9

Key points

1. $E_{cell} = E_{cathode} - E_{anode}$ (right minus left) is used in every cell numerical of the UE23/UE24 papers (Q2), so this convention is examined every time. ESA ★★★ S001·9 · ESA Jul-23a 2c · ESA Dec-23 2c · ESA Jun-24b 2a
2. E_{cell} depends on the concentrations, the temperature and the partial pressures of gases; E°_{cell} fixes them at 1 M, 1 atm and 298 K. ESA ★★ S001·9 · T001 00:14:43
3. E_{cell} is the driving force of the cell reaction: a system moves toward minimum free energy, and $\Delta G = -nFE_{cell}$. ESA ★★★ S001·9 · T001 00:16:15
4. E°_{cell} is intensive: it does not change when the cell reaction is multiplied by a number, but ΔG° does (through n). ESA ★★ added for clarity S001·9

Why?

Why is $\Delta G = -nFE$?
The maximum work a cell can do equals the total charge times the energy per unit charge: $W_{max} = nF \times E$; the maximum (non-expansion) work equals $-\Delta G$. Derived in class 002.
T001 00:20:11 · T001 00:21:06

Why subtract reduction potentials instead of adding an oxidation potential?
Both are equivalent: $E^\circ_{ox}(\text{anode}) = -E^\circ_{red}(\text{anode})$, so $E^\circ_{red,cathode} + E^\circ_{ox,anode} = E^\circ_{cathode} - E^\circ_{anode}$. The slides tabulate reduction potentials, so subtract.
added for clarity S001·9 · S001·10

Exam traps

Exam trap. Never double-flip signs: in $E^\circ_{cathode} - E^\circ_{anode}$ use the tabulated reduction potentials as they are. $0.34 - (-0.76) = 1.10$ V, not $0.34 - 0.76$. added for clarity S001·9

Exam trap. Do not multiply E° by stoichiometric coefficients when balancing (e.g. $\mathbf{2Al + 3Cu^{2+}}$); only n in $\Delta G = -nFE$ changes. ADDED FOR CLARITY ESA Jan-25 s1 2a

Exam trap. $\Delta G = -nFE$ gives joules per mole of reaction; divide by 1000 for kJ. n is the number of electrons in the balanced overall reaction (6 for Fe/Au, 2 for Daniell). ADDED FOR CLARITY S001·9

5 The electrochemical series and its uses

Definitions

Electrochemical series. In order to predict the electrochemical behaviour of an electrode–electrolyte system, elements are arranged in the order of their standard reduction potentials. This arrangement is known as the electrochemical series. S001·10 · T001 00:17:34

Statements in the class material

> *"A negative value indicates oxidation tendency while a positive value indicates a reduction tendency."* S001·10

Key points

1. The series runs from Li (-3.03 V, oxidised most easily) to Au ($+1.50$ V on the slide, reduced most easily). Hydrogen (0.00 V) is the reference. ESA ★★ S001·10 · T001 00:18:21
2. Of two electrodes, the one with the lower (more negative) E° is the anode, and E°_{cell} is the gap between them on the series. ESA ★★★ added for clarity S001·10
3. A metal above hydrogen (negative E°, e.g. Zn) liberates $\mathbf{H_2}$ from dilute acid; one below hydrogen (Cu, Ag, Au) does not. ESA ★★ added for clarity S001·10
4. A metal displaces from solution the ions of any metal below it in the series (Zn + $\mathbf{Cu^{2+}}$ → $\mathbf{Zn^{2+}}$ + Cu). ESA ★★ added for clarity S001·10
5. Corrosion: when two metals touch in a moist environment, the one with the lower E° becomes the anode and corrodes (used again in the corrosion classes 006–011). ESA ★★ added for clarity ESA Jul-23a 4d

Why?

Why does a negative E° mean an oxidation tendency?
Coupled with the hydrogen electrode, such a metal gives $E^\circ_{cell} > 0$ only as the anode, i.e. when it is oxidised; Li (-3.03 V) oxidises very easily while $\mathbf{Cu^{2+}}$ ($+0.34$ V) is easily reduced.
S001·10 · T001 00:18:05

Exam traps

Exam trap. The slide prints $E^\circ(\mathbf{Au^{3+}/Au}) = +1.50$ V, but the ESA Fe/Au problems give 1.52 V. Always use the value printed in the question. S001·10 · ESA Dec-23 2c · ESA Jun-24b 2a

Exam trap. The slide's right-hand arrow reads "Metal Oxidizing Activity Increasing" downwards; it is the *ion* (oxidant) whose oxidising power increases downwards. The metal's tendency to be oxidised increases *upwards*. added for clarity S001·10

Exam trap. A tin coating protects iron only while it is intact: Sn (-0.14 V) is above Fe (-0.44 V) in E°, so at a scratch iron becomes the anode and corrodes faster. ADDED FOR CLARITY ESA Jul-23a 4d

★ Past-paper questions

Ten unique ESA papers were checked (UE22CY151B Jul 2023 a/b, Dec 2023, May 2024, Dec 2024; UE23CY151A/B Jan 2024 set 2, Jun 2024 a/b, Dec 2024; UE24CY151A Jan 2025 sets 1 and 2; three 2024 files are one paper). Directly on this class: Jan 2025 set 2 Q2.b(i), 3 marks (double layer, case I). The electrochemical series is applied in Jul 2023 Q4.d(i) (zinc versus tin coating; primary class 010). $E_{cell} = E_{cathode} - E_{anode}$ is used in every UE23/UE24 Q2 cell numerical (Jul-23a 2.c, Dec-23 2.c, Jun-24b 2.a and others; primary class 006), so the convention is rated ★★★. The galvanic/electrolytic comparison, cell notation as a stand-alone question, the standard-electrode-potential definition and $\Delta G = -nFE$ as a separate question were never asked.

2 · Electrode potential and its origin: the Helmholtz double layer

ESA Jan 2025 (3 marks) UE24CY151A set 2, Q2.b(i) · solved in W2.1

When a metal M is in contact with solution containing its ions $\mathbf{M^{n+}}$ and if ionization is faster than deposition, explain the formation of the electrical double layer at the interface of metal and solution.

4 · Cell potential, standard cell potential and free energy

ESA Jul 2023 (5 marks) UE22CY151B Q2.c (E°cell part) · solved in W4.3

A silver electrode containing 0.15 M silver nitrate and copper electrode containing 0.19 M copper sulphate were coupled using salt bridge. Write the cell representation, half cell reactions, calculate E°_{cell}, E_{cell} at 25 °C. (Primary session 006; the E°_{cell} step uses this class.)

ESA Dec 2023 / Jun 2024 (5 marks) UE22CY151B Q2.c; UE23CY151B Q2.a (E°cell part) · solved in W4.4

$\mathbf{Fe/Fe^{2+}(0.1\,M)\,||\,Au^{3+}(0.5\,M)/Au}$: half-cell reactions, E°_{cell} and E_{cell} at 298 K ($E^{\circ}_{Au^{3+}/Au} = 1.52$ V, $E^{\circ}_{Fe^{2+}/Fe} = -0.44$ V). Dec-23: 5 marks; Jun-24b: 6 marks. (Primary session 006.)

5 · The electrochemical series and its uses

ESA Jul 2023 (2 marks) UE22CY151B Q4.d(i) · solved in W5.2

Galvanization and tinning are two popular methods of protecting a metal by coating with another metal. If the reduction potentials of iron, zinc and tin are −0.44 V, −0.76 V and −0.14 V respectively, then comment on the situation: corrosion of iron would be observed in case a section of the protecting layer wears off. Briefly discuss the reason. (4.d is 4 marks with part ii.)